CSCI 2114: Data Structures Syllabus

Fall 2026

Oklahoma City University, Petree College of Arts & Sciences, Computer Science

1. Instructor Information and Course Material

Professor: Dr. Tashfeen, Ahmad (MSB 219B, tashfeen@okcu.edu[1])
Office Hours: Mondays, Wednesdays during 10–11 am and 5–6:30 pm or by appointment.[2]
Textbook: Cormen, et al. *Introduction to Algorithms* 3rd Ed. (2009).[3]
Time & Location: Monday, Wednesday at 2:30–3:45 pm in MSB 201.

2. Tentative Schedule

Week	Friday	Topic	Event	(%)
1	Aug 28	Arrays, Search	Homework 1	5
2	Sep 4	Asymptotic Analysis	Homework 2	5
3	Sep 11	Sorting		
4	Sep 18	Sorting	Homework 3	5
5	Sep 25	Array Lists, Linked Lists		
6	Oct 2	Stacks, Queues, Heaps	Homework 4	10
7	Oct 9	Stacks, Queues, Heaps		
8	Oct 16	Review	Midterm	20
9	Oct 23		Fall Break	
10	Oct 30	Trees		
11	Nov 6	Trees	Homework 5	10
12	Nov 13	Hash Maps, Graphs		
13	Nov 20	Hash Maps, Graphs	Homework 6	10
14	Nov 27		Thanksgiving	
15	Dec 4	https://projecteuler.net/	Homework 7	10
16	Dec 11	Review		
17	Dec 18	Review	Final	20

Table 1 — 5% is for quizzes and/or class participation.

3. Grading Criteria

When accepted, work that is $n < 3$ days late will receive no more than $100(1 - 0.1n)\%$ of credit (10% deduction per day). For certain work, late submissions will not be accepted. Any submission that is not easy to read will receive a zero.

Total t %	Grade	Total t %	Grade	Total t %	Grade
		$t \geq 93$	A	$93 > t \geq 90$	A^-
$90 > t \geq 85$	B^+	$85 > t \geq 82$	B	$82 > t \geq 80$	B^-
$80 > t \geq 75$	C^+	$75 > t \geq 72$	C	$72 > t \geq 70$	C^-
$70 > t \geq 65$	D^+	$65 > t \geq 62$	D	$62 > t \geq 60$	D^-
$60 > t$	F				

Table 2 — A grade of Incomplete ("I") will only be assigned in case of documented extenuating circumstances. The "I" will be removed in accordance with university policy stated in the online undergraduate catalogue. Please also see subsection 4.4.

[1] Prefix [XXXX1234] followed by a space to all emails sent to the instructor, e. g., "[CSCI2114] Help with Homework 7."
[2] Zoom for a virtual meeting.
[3] A copy should be available at the library.

4. Policies & Resources

4.1. General Academic Guidelines. For an up-to-date version of the guidelines please visit the university sharepoint, see the OCU course schedule website for the finals week schedule and visit the online classroom.

4.2. Course Description. This course examines how to organise data so that it can be stored compactly and retrieved swiftly. It covers linear containers such as arrays and lists, hierarchical forms like trees and heaps, abstract types including stacks, queues and hash tables, and algorithmic techniques for sorting, searching and traversing graphs. All concepts are reinforced with from-scratch implementations in plain Java.

4.3. Course Learning Outcomes.

- Describe the fundamental principles of data organisation and the trade-offs of different structures for storage and retrieval.
- Derive the theoretical underpinnings of key data-structuring techniques, including complexity analysis of operations.
- Implement the covered structures and associated algorithms from first principles using only core Java (no external libraries).
- Evaluate the performance of different structures and algorithms using appropriate metrics and visual diagnostics, and communicate the results clearly.

4.4. Academic Integrity. The cheating rule for this class is simple: *don't turn in anything you did not understand*. I encourage you to get help and exhaust your resources. Though, if you turn in something (or answer a question with something you do not understand; can not explain) and I unsuccessfully ask you to explain your work, that will result in an *automatic F in the course* and disciplinary action will be taken. If you've been asked to demonstrate your work to the professor then you'll need to do so in his office hours or make an appointment. If you've been asked to demonstrate your work and you fail to do so, you will receive a zero in the assignment. For more, read the *Academic Honesty* section of the courses' catalogue.

4.4.1. *Artificial Intelligence (AI).* Help obtained by an AI, e. g., Large Language Model chat agent should be documented and turned in with the work. You may use AI to understand a general concept but it should *never* write or solve anything for you. A student who abuses AI will forfeit the right to use it for the remainder of the class.

4.5. Religious Accommodation. Oklahoma City University seeks to be supportive of religious observance among the members of our diverse campus community and to be as accommodating as possible. Students should discuss with their instructor at the beginning of the semester forms of religious observance (dress, fasting, specific prayer times) that may affect their full participation in the course. Students should also compare the class schedule to their own religious calendar to determine if there will be any class days in which the student expects to be absent due to the observance of a religious holiday. *Students must notify the instructor, in writing, of the expected absence within the first two weeks of the semester.* The instructor will then work with the student to develop a plan to reschedule any exams, assignments, or course activities for that day. The instructor, at his/her own discretion, will make reasonable accommodations wherever possible. Students should recognize that there may be some course aspects that cannot be rescheduled or accommodated, and it will therefore rest upon the student to determine whether they wish to remain enrolled in the course or have their grade potentially affected.

4.6. Mission Statement. The Petree College of Arts and Sciences provides a supportive, student-centered learning environment. Through an interdisciplinary curriculum that promotes active learning and individualized instruction, students are encouraged to think broadly, to find their passion, to collaborate with others, and to connect with the broader community.

4.7. Grook.

Put up in a place
where it's easy to see
the cryptic admonishment
T. T. T.

When you feel how depressingly
slowly you climb,
it's well to remember that
Things Take Time

——*Piet Hein (1970)*